

Cover Letter : Evidence Synthesis Article

Why do Birds use Green Nest Material? A systematic review and meta-analysis of experiments

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Nest-building is a widespread behaviour across the animal kingdom, from arthropods to mammals ^[1,2]. Serving as a shelter to eggs, offspring and parents, nests play a key role in the reproductive cycle and are subject to natural selection. Among animals that build nests, birds in particular provide some of the most striking examples of complex nest structures, which function as extended phenotypes ^[3], acted upon by sexual selection. There has been increasing interest in nests over the past decade ^[4,5]. For example, a title-abstract-keyword search for “nests” on Web of Science yields over 32,000 articles in the field of biology alone, highlighting the widespread interest in this fascinating structure. One such intriguing behaviour of nest construction by birds is the addition of fresh, often aromatic, green plant material to nests ^[6]. For decades, researchers have debated whether this behaviour reflects sexual signalling, parasite reduction, or has other adaptive functions. Despite the accumulation of experimental studies for over 40 years, the adaptive value of green nest material remains unclear.

To address this knowledge gap, we have conducted a **pre-registered** ^[7] **systematic review and meta-analysis** synthesising all experimental evidence on the fitness consequences of this behaviour. We find that green nest material has an overall positive effect on fitness, though the mechanism of this benefit remains unclear. By testing the

major proposed hypotheses, we challenge long-standing assumptions that aromatic volatiles are the primary adaptive mechanisms ^[8] and instead highlight the critical importance of experimental design in understanding outcomes. This meta-analysis reshapes the field by providing the first quantitative evidence of green nest material's adaptive significance, identifying key methodological biases, and reframing the questions that should guide future research. The synthesis is especially timely given the rapid growth of evidence in this area and the increasing number of experiments being conducted to answer these questions.

Given the conceptual importance of nest-building across taxa and the accessibility of birds as a model system, our article will be of broad interest. Our team combines expertise in meta-analytical methods, avian ecology, field experimentation on green nest material behaviour, and extended phenotypes across taxa (including birds and arthropods), enabling us to situate our findings within the wider context of sexual selection, parental care, and animal architecture. This comprehensive approach ensures our work will appeal to ornithologists, evolutionary biologists, behavioural ecologists, and readers interested in adaptive structures across the animal kingdom.

Best regards,

Shreya Dimri,
On behalf of all co-authors.

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